Vignette 1

The Ministry of Magic decided to institute a price ceiling to allow many more to afford the treat. Find an appropriate price for the price ceiling and use a graph to evaluate the welfare effects your policy had on the market.

The supply and demand curves for butterbeer can be represented by the following equations.

$$P = 23 - \frac{1}{6}Q$$

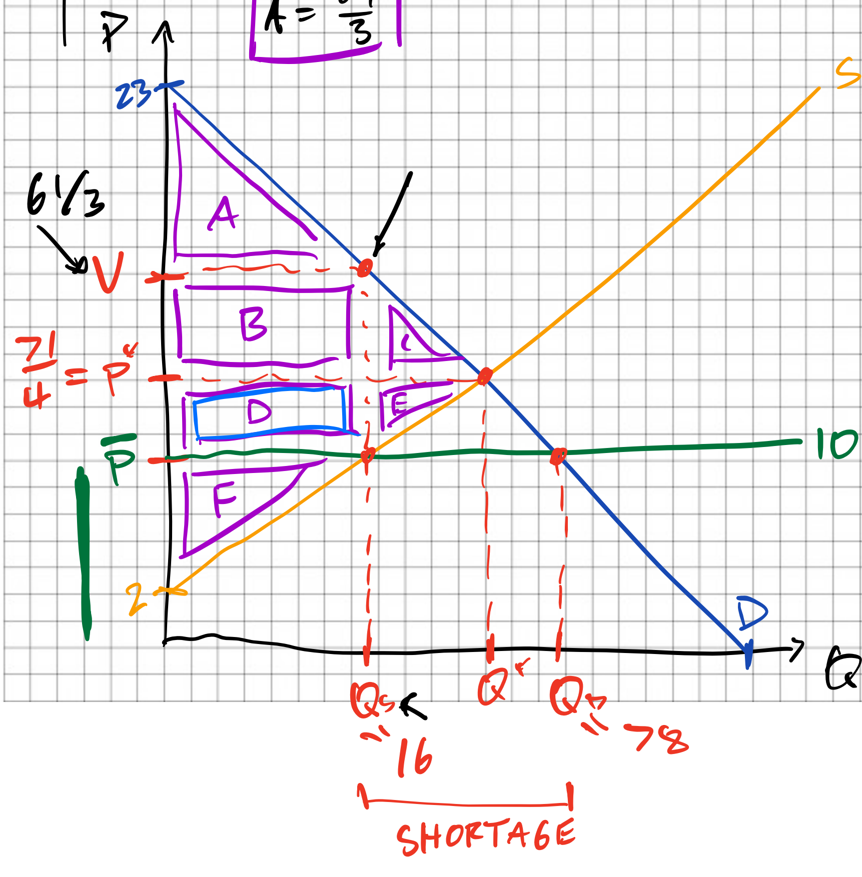
$$P = 2 + \frac{1}{2}Q$$

$V = 23 - \frac{1}{6}\cdot 16$
$= 23 - \frac{8}{3}$
$= \frac{69 - 8}{3}$
$V = \frac{61}{3}$

$A = h \times b \times \frac{1}{2}$
$A = \left(23 - \frac{61}{3}\right) \times 16 \times \frac{1}{2}$
$A = \frac{69 - 61}{3} \times 8$
$\boxed{A = \frac{64}{3}}$

$23 - \frac{1}{6}Q = 2 + \frac{1}{2}Q$
$21 = \left(\frac{1}{6} + \frac{3}{6}\right)Q = \frac{2}{3}Q$
$Q^* = 21 \cdot \frac{3}{2} \Rightarrow \boxed{Q^* = \frac{63}{2}}$

$P^* = 2 + \frac{1}{2}\cdot\left(\frac{63}{2}\right)$
$P^* = \frac{8}{4} + \frac{63}{4}$
$\boxed{P^* = \frac{71}{4}} \approx 18$



$10 = 2 + \frac{1}{2}Q_S$
$8 = \frac{1}{2}Q_S$
$\boxed{Q_S = 16}$

$10 = 23 - \frac{1}{6}Q_D$
$\frac{1}{6}Q_S = 13$
$Q_S = 13 \cdot 6 = 60 + 18$
$\boxed{Q_S = 78}$

PRE PRICE CEILING

CS | A + B + C

PS | D + E + F

POST PRICE CEILING

CS | A + B + D

PS | F

DWL | C + E

The policy lowered price, created a shortage, and generated DWL.

It also transfered welfare D from PS → CS.

We would expect CS to increase.

Vignette 2

The Ministry was in some financial trouble after their considerable expenditures duing the war, and decided to impose a tax. Use a graph to evaluate the welfare effects the policy had on the market.

The supply and demand curves for butterbeer can be represented by the following equations.

$$P_B = 23 - \frac{1}{6}Q$$

$$P_S = 2 + \frac{1}{2}Q$$

$P_B = 23 - \frac{1}{6}\cdot 24$

$P_B = 23 - 4$

$\boxed{P_B = 14}$

$P_B = P_S - 5$

$\boxed{P_S = 14}$

$P_B = P_S + TAX$

$Q_S = Q_D$

$TAX = 5$

$19 = P_S + 5$

$P_B = P_S + 5$

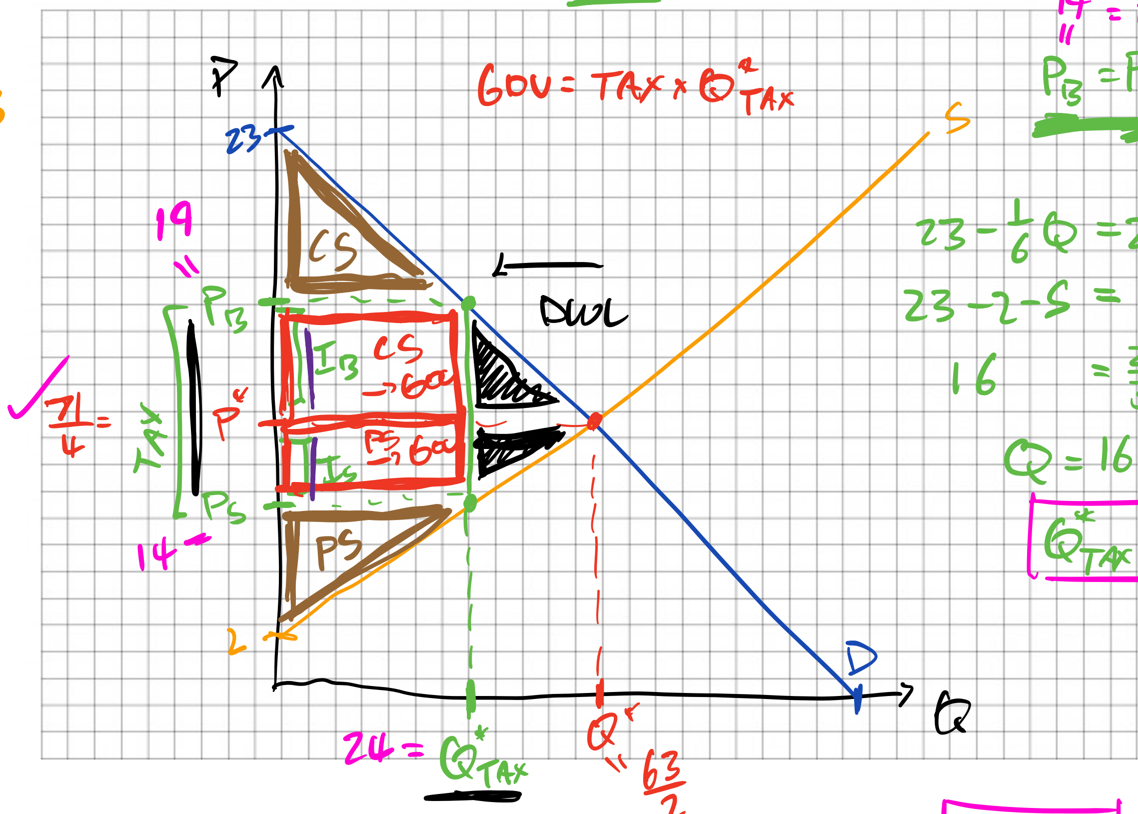
$23 - \frac{1}{6}Q = 2 + \frac{1}{2}Q + 5$

$23 - 2 - 5 = \left(\frac{1}{6} + \frac{3}{6}\right)Q$

$16 = \frac{2}{3}Q$

$Q = 16 \cdot \frac{3}{2} = 8 \times 3$

$\boxed{Q^*_{TAX} = 24}$



$I_B = P_B - P^* = 19 - \frac{71}{4} = \frac{80-4}{4} - \frac{71}{4} = \frac{76-71}{4} = \boxed{\frac{5}{4} = I_B}$

$I_S = 5 - \frac{5}{4} = \frac{20-5}{4} = \boxed{\frac{15}{4} = I_S}$

After the tax, quantity exchanged decreases. This leads to DWL.
We also generate government revenue from both PS & CS.
But the incidence of tax falls mainly on sellers.